

****CLIMATE CHANGE: EVIDENCE & CAUSES****

****UPDATE 2020****

****An overview from the Royal Society and the US National Academy of Sciences****

****FOREWORD****

Climate change is one of the defining issues of our time. It is now more certain than ever, based on many lines of evidence, that humans are changing Earth's climate. The atmosphere and oceans have warmed, accompanied by sea level rise, a strong decline in Arctic sea ice, and other climate-related changes. The impacts of climate change on people and nature are increasingly apparent. Unprecedented flooding, heat waves, and wildfires have cost billions in damages. Habitats are undergoing rapid shifts in response to changing temperatures and precipitation patterns.

The Royal Society and the US National Academy of Sciences, with their similar missions to promote the use of science to benefit society and to inform critical policy debates, produced the original *Climate Change: Evidence and Causes* in 2014. It was written and reviewed by a UK-US team of leading climate scientists. This new edition, prepared by the same author team, has been updated with the most recent climate data and scientific analyses, all of which reinforce our understanding of human-caused climate change.

The evidence is clear. However, due to the nature of science, not every detail is ever totally settled or certain. Nor has every pertinent question yet been answered. Scientific evidence continues to be gathered around the world. Some things have become clearer, and new insights have emerged. For example, the period of slower warming during the 2000s and early 2010s has ended with a dramatic jump to warmer temperatures between 2014 and 2015. Antarctic sea ice extent, which had been increasing, began to decline in 2014, reaching a record low in 2017 that has persisted. These and other recent observations have been woven into the discussions of the questions addressed in this booklet.

Scientific information is a vital component for society to make informed decisions about how to reduce the magnitude of climate change and how to adapt to its

impacts. This booklet serves as a key reference document for decision-makers, policymakers, educators, and others seeking authoritative answers about the current state of climate-change science.

We are grateful that six years ago, under the leadership of Dr. Ralph J. Cicerone, former President of the National Academy of Sciences, and Sir Paul Nurse, former President of the Royal Society, these two organizations partnered to produce a high-level overview of climate change science. As current Presidents of these organizations, we are pleased to offer an update to this key reference, supported by the generosity of the Cicerone Family.

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****SUMMARY****

Greenhouse gases such as carbon dioxide (CO₂) absorb heat (infrared radiation) emitted from Earth's surface. Increases in the atmospheric concentrations of these gases cause Earth to warm by trapping more of this heat. Human activities—especially the burning of fossil fuels since the start of the Industrial Revolution—have increased atmospheric CO₂ concentrations by more than 40%, with over half the increase occurring since 1970. Since 1900, the global average surface temperature has increased by about 1 °C (1.8 °F). This has been accompanied by warming of the ocean, a rise in sea level, a strong decline in Arctic sea ice, widespread increases in the frequency and intensity of heatwaves, and many other associated climate effects. Much of this warming has occurred in the last five decades. Detailed analyses have shown that the warming during this period is mainly a result of the increased concentrations of CO₂ and other greenhouse gases. Continued emissions of these gases will cause further climate change, including substantial increases in global average surface temperature and important changes in regional climate. The magnitude and timing of these changes will depend on many factors, and slowdowns and accelerations in warming lasting a decade or

more will continue to occur. However, long-term climate change over many decades will depend mainly on the total amount of CO₂ and other greenhouse gases emitted as a result of human activities.

****CLIMATE CHANGE Q&A****

****1. Is the climate warming?****

Yes. Earth's average surface air temperature has increased by about 1 °C (1.8 °F) since 1900, with over half of the increase occurring since the mid-1970s. A wide range of other observations, such as reduced Arctic sea ice extent and increased ocean heat content, together provide incontrovertible evidence of planetary-scale warming.

The clearest evidence for surface warming comes from widespread thermometer records that, in some places, extend back to the late 19th century. Today,

temperatures are monitored at many thousands of locations, over both the land and ocean surface. Indirect estimates of temperature change from such sources as tree rings and ice cores help to place recent temperature changes in the context of the past. In terms of the average surface temperature of Earth, these indirect estimates show that 1989 to 2019 was very likely the warmest 30-year period in more than 800 years; the most recent decade, 2010-2019, is the warmest decade in the instrumental record so far (since 1890).

****2. How do scientists know that recent climate change is largely caused by human activities?***

Scientists know that recent climate change is largely caused by human activities from an understanding of basic physics, comparing observations with models, and fingerprinting the detailed patterns of climate change caused by different human and natural influences.

Since the mid-1800s, scientists have known that CO₂ is one of the main greenhouse gases of importance to Earth's energy balance. Direct measurements of CO₂ in the atmosphere and in air trapped in ice show that atmospheric CO₂ increased by more

than 40% from 1800 to 2019. Measurements of different forms of carbon (isotopes) reveal that this increase is due to human activities. Other greenhouse gases (notably methane and nitrous oxide) are also increasing as a consequence of human activities. The observed global surface temperature rise since 1900 is consistent with detailed calculations of the impacts of the observed increase in atmospheric greenhouse gases (and other human-induced changes) on Earth's energy balance.

Different influences on climate have different signatures in climate records. These unique fingerprints are easier to see by probing beyond a single number (such as the average temperature of Earth's surface), and by looking instead at the geographical and seasonal patterns of climate change. The observed patterns of surface warming, temperature changes through the atmosphere, increases in ocean heat content, increases in atmospheric moisture, sea level rise, and increased melting of land and sea ice also match the patterns scientists expect to see due to human activities.

The expected changes in climate are based on our understanding of how greenhouse gases trap heat. Both this fundamental understanding of the physics of

greenhouse gases and pattern-based fingerprint studies show that natural causes alone are inadequate to explain the recent observed changes in climate. Natural causes include variations in the Sun's output and in Earth's orbit around the Sun, volcanic eruptions, and internal fluctuations in the climate system (such as El Niño and La Niña). Calculations using climate models have been used to simulate what would have happened to global temperatures if only natural factors were influencing the climate system. These simulations yield little surface warming, or even a slight cooling, over the 20th century and into the 21st. Only when models include human influences on the composition of the atmosphere are the resulting temperature changes consistent with observed changes.

****3. CO₂ is already in the atmosphere naturally, so why are emissions from human activity significant?****

Human activities have significantly disturbed the natural carbon cycle by extracting long-buried fossil fuels and burning them for energy, thus releasing CO₂ to the atmosphere.

In nature, CO₂ is exchanged continually between the atmosphere, plants, and animals through photosynthesis, respiration, and decomposition, and between the atmosphere and ocean through gas exchange. A very small amount of CO₂ (roughly 1% of the emission rate from fossil fuel combustion) is also emitted in volcanic eruptions. This is balanced by an equivalent amount that is removed by chemical weathering of rocks.

The CO₂ level in 2019 was more than 40% higher than it was in the 19th century. Most of this CO₂ increase has taken place since 1970, about the time when global energy consumption accelerated. Measured decreases in the fraction of other forms of carbon (the isotopes ¹³C and ¹⁴C) and a small decrease in atmospheric oxygen concentration (observations of which have been available since 1990) show that the rise in CO₂ is largely from combustion of fossil fuels (which have low ¹³C fractions and no ¹⁴C). Deforestation and other land use changes have also released carbon from the biosphere (living world) where it normally resides for decades to centuries. The additional CO₂ from fossil fuel burning and deforestation has disturbed the balance of the carbon cycle, because the natural processes that could restore the balance are too slow compared to the rates at which human activities

are adding CO₂ to the atmosphere. As a result, a substantial fraction of the CO₂ emitted from human activities accumulates in the atmosphere, where some of it will remain not just for decades or centuries, but for thousands of years.

Comparison with the CO₂ levels measured in air extracted from ice cores indicates that the current concentrations are substantially higher than they have been in at least 800,000 years.

****4. What role has the Sun played in climate change in recent decades?****

The Sun provides the primary source of energy driving Earth's climate system, but its variations have played very little role in the climate changes observed in recent decades. Direct satellite measurements since the late 1970s show no net increase in the Sun's output, while at the same time global surface temperatures have increased.

For periods before the onset of satellite measurements, knowledge about solar changes is less certain because the changes are inferred from indirect sources—including the number of sunspots and the abundance of certain forms (isotopes) of carbon or beryllium atoms, whose production rates in Earth's atmosphere are influenced by variations in the Sun. There is evidence that the 11-year solar cycle, during which the Sun's energy output varies by roughly 0.1%, can influence ozone concentrations, temperatures, and winds in the stratosphere (the layer in the atmosphere above the troposphere, typically from 12 to 50 km above Earth's surface, depending on latitude and season). These stratospheric changes may have a small effect on surface climate over the 11-year cycle. However, the available evidence does not indicate pronounced long-term changes in the Sun's output over the past century, during which time human-induced increases in CO₂ concentrations have been the dominant influence on the long-term global surface temperature increase. Further evidence that current warming is not a result of solar changes can be found in the temperature trends at different altitudes in the atmosphere.

****5. What do changes in the vertical structure of atmospheric temperature—from the surface up to the stratosphere—tell us about the causes of recent climate change?****

The observed warming in the lower atmosphere and cooling in the upper atmosphere provide us with key insights into the underlying causes of climate change and reveal that natural factors alone cannot explain the observed changes.

In the early 1960s, results from mathematical/physical models of the climate system first showed that human-induced increases in CO₂ would be expected to lead to gradual warming of the lower atmosphere (the troposphere) and cooling of higher levels of the atmosphere (the stratosphere). In contrast, increases in the Sun's output would warm both the troposphere and the full vertical extent of the stratosphere. At that time, there was insufficient observational data to test this prediction, but temperature measurements from weather balloons and satellites have since confirmed these early forecasts. It is now known that the observed pattern of tropospheric warming and stratospheric cooling over the past 40 years is broadly consistent with computer model simulations that include increases in CO₂ and decreases in stratospheric ozone, each caused by human activities. The

observed pattern is not consistent with purely natural changes in the Sun's energy output, volcanic activity, or natural climate variations such as El Niño and La Niña.

****6. Climate is always changing. Why is climate change of concern now?****

All major climate changes, including natural ones, are disruptive. Past climate changes led to extinction of many species, population migrations, and pronounced changes in the land surface and ocean circulation. The speed of the current climate change is faster than most of the past events, making it more difficult for human societies and the natural world to adapt.

The largest global-scale climate variations in Earth's recent geological past are the ice age cycles, which are cold glacial periods followed by shorter warm periods. The last few of these natural cycles have recurred roughly every 100,000 years. They are mainly paced by slow changes in Earth's orbit, which alter the way the Sun's energy is distributed with latitude and by season on Earth. These orbital changes are very small over the last several hundred years, and alone are not sufficient to cause the observed magnitude of change in temperature since the Industrial Revolution, nor to act on the whole Earth. On ice-age timescales, these

gradual orbital variations have led to changes in the extent of ice sheets and in the abundance of CO₂ and other greenhouse gases, which in turn have amplified the initial temperature change.

Recent estimates of the increase in global average temperature since the end of the last ice age are 4 to 5 °C (7 to 9 °F). That change occurred over a period of about 7,000 years, starting 18,000 years ago. CO₂ has risen more than 40% in just the past 200 years, much of this since the 1970s, contributing to human alteration of the planet's energy budget that has so far warmed Earth by about 1 °C (1.8 °F). If the rise in CO₂ continues unchecked, warming of the same magnitude as the increase out of the ice age can be expected by the end of this century or soon after. This speed of warming is more than ten times that at the end of an ice age, the fastest known natural sustained change on a global scale.

****7. Is the current level of atmospheric CO₂ concentration unprecedented in Earth's history?****

The present level of atmospheric CO₂ concentration is almost certainly unprecedented in the past million years, during which time modern humans evolved and societies developed. The atmospheric CO₂ concentration was however higher in Earth's more distant past (many millions of years ago), at which time palaeoclimatic and geological data indicate that temperatures and sea levels were also higher than they are today.

Measurements of air in ice cores show that for the past 800,000 years up until the 20th century, the atmospheric CO₂ concentration stayed within the range 170 to 300 parts per million (ppm), making the recent rapid rise to more than 400 ppm over 200 years particularly remarkable. During the glacial cycles of the past 800,000 years, both CO₂ and methane have acted as important amplifiers of the climate changes triggered by variations in Earth's orbit around the Sun. As Earth warmed from the last ice age, temperature and CO₂ started to rise at approximately the same time and continued to rise in tandem from about 18,000 to 11,000 years ago. Changes in ocean temperature, circulation, chemistry, and biology caused CO₂ to be released to the atmosphere, which combined with other feedbacks to push Earth into an even warmer state.

For earlier geological times, CO₂ concentrations and temperatures have been inferred from less direct methods. Those suggest that the concentration of CO₂ last approached 400 ppm about 3 to 5 million years ago, a period when global average surface temperature is estimated to have been about 2 to 3.5 °C higher than in the pre-industrial period. At 50 million years ago, CO₂ may have reached 1000 ppm, and global average temperature was probably about 10 °C warmer than today. Under those conditions, Earth had little ice, and sea level was at least 60 metres higher than current levels.

****8. Is there a point at which adding more CO₂ will not cause further warming?****

No. Adding more CO₂ to the atmosphere will cause surface temperatures to continue to increase. As the atmospheric concentrations of CO₂ increase, the addition of extra CO₂ becomes progressively less effective at trapping Earth's energy, but surface temperature will still rise.

Our understanding of the physics by which CO₂ affects Earth's energy balance is confirmed by laboratory measurements, as well as by detailed satellite and surface observations of the emission and absorption of infrared energy by the atmosphere. Greenhouse gases absorb some of the infrared energy that Earth emits in so-called bands of stronger absorption that occur at certain wavelengths. Different gases absorb energy at different wavelengths. CO₂ has its strongest heat-trapping band centered at a wavelength of 15 micrometers (millionths of a meter), with absorption that spreads out a few micrometers on either side. There are also many weaker absorption bands. As CO₂ concentrations increase, the absorption at the centre of the strong band is already so intense that it plays little role in causing additional warming. However, more energy is absorbed in the weaker bands and away from the centre of the strong band, causing the surface and lower atmosphere to warm further.

****9. Does the rate of warming vary from one decade to another?****

Yes. The observed warming rate has varied from year to year, decade to decade, and place to place, as is expected from our understanding of the climate system. These shorter-term variations are mostly due to natural causes, and do not contradict our fundamental understanding that the long-term warming trend is primarily due to human-induced changes in the atmospheric levels of CO₂ and other greenhouse gases.

Even as CO₂ is rising steadily in the atmosphere, leading to gradual warming of Earth's surface, many natural factors are modulating this long-term warming. Large volcanic eruptions increase the number of small particles in the stratosphere. These particles reflect sunlight, leading to short-term surface cooling lasting typically two to three years, followed by a slow recovery. Ocean circulation and mixing vary naturally on many time scales, causing variations in sea surface temperatures as well as changes in the rate at which heat is transported to greater depths. For example, the tropical Pacific swings between warm El Niño and cooler La Niña events on timescales of two to seven years. Scientists study many different types of climate variations, such as those on decadal and multi-decadal timescales in the Pacific and North Atlantic Oceans. Each type of variation has its own unique characteristics. These oceanic variations are associated with significant regional

and global shifts in temperature and rainfall patterns that are evident in the observations.

Warming from decade to decade can also be affected by human factors such as variations in emissions of greenhouse gases and aerosols (airborne particles that can have both warming and cooling effects) from coal-fired power plants and other pollution sources.

These variations in the temperature trend are clearly evident in the observed temperature record. Short-term natural climate variations could also affect the long-term human-induced climate change signal and vice-versa, because climate variations on different space and timescales can interact with one another. It is partly for this reason that climate change projections are made using climate models that can account for many different types of climate variations and their interactions. Reliable inferences about human-induced climate change must be made with a longer view, using records that cover many decades.

****10. Did the slowdown of warming during the 2000s to early 2010s mean that climate change is no longer happening?****

No. After the very warm year 1998 that followed the strong 1997-98 El Niño, the increase in average surface temperature slowed relative to the previous decade of rapid temperature increases. Despite the slower rate of warming, the 2000s were warmer than the 1990s. The limited period of slower warming ended with a dramatic jump to warmer temperatures between 2014 and 2015, with all the years from 2015-2019 warmer than any preceding year in the instrumental record. A short-term slowdown in the warming of Earth's surface does not invalidate our understanding of long-term changes in global temperature arising from human-induced changes in greenhouse gases.

Decades of slow warming as well as decades of accelerated warming occur naturally in the climate system. Decades that are cold or warm compared to the long-term trend are seen in the observations of the past 150 years and are also captured by climate models. Because the atmosphere stores very little heat, surface temperatures can be rapidly affected by heat uptake elsewhere in the climate

system and by changes in external influences on climate (such as particles formed from material lofted high into the atmosphere from volcanic eruptions).

More than 90% of the heat added to the Earth system in recent decades has been absorbed by the oceans and penetrates only slowly into deep water. A faster rate of heat penetration into the deeper ocean will slow the warming seen at the surface and in the atmosphere, but by itself it will not change the long-term warming that will occur from a given amount of CO₂. For example, recent studies show that some heat comes out of the ocean into the atmosphere during warm El Niño events, and more heat penetrates to ocean depths in cold La Niñas. Such changes occur repeatedly over timescales of decades and longer. An example is the major El Niño event in 1997–98 when the globally averaged air temperature soared to the highest level in the 20th century as the ocean lost heat to the atmosphere, mainly by evaporation.

****11. If the world is warming, why are some winters and summers still very cold?****

Global warming is a long-term trend, but that does not mean that every year will be warmer than the previous one. Day-to-day and year-to-year changes in weather patterns will continue to produce some unusually cold days and nights, winters, and summers, even as the climate warms.

Climate change means not only changes in globally averaged surface temperature but also changes in atmospheric circulation, in the size and patterns of natural climate variations, and in local weather. La Niña events shift weather patterns so that some regions are made wetter, and wet summers are generally cooler. Stronger winds from polar regions can contribute to an occasional colder winter. In a similar way, the persistence of one phase of an atmospheric circulation pattern known as the North Atlantic Oscillation has contributed to several recent cold winters in Europe, eastern North America, and northern Asia.

Atmospheric and ocean circulation patterns will evolve as Earth warms and will influence storm tracks and many other aspects of the weather. Global warming tilts the odds in favour of more warm days and seasons and fewer cold days and seasons. For example, across the continental United States in the 1960s, there were more daily record low temperatures than record highs, but in the 2000s, there were

more than twice as many record highs as record lows. Another important example of tilting the odds is that over recent decades, heatwaves have increased in frequency in large parts of Europe, Asia, South America, and Australia. Marine heatwaves are also increasing.

****12. Why is Arctic sea ice decreasing while Antarctic sea ice has changed little?****

Sea ice extent is affected by winds and ocean currents as well as temperature. Sea ice in the partly-enclosed Arctic Ocean seems to be responding directly to warming, while changes in winds and in the ocean seem to be dominating the patterns of climate and sea ice change in the ocean around Antarctica.

Some differences in seasonal sea ice extent between the Arctic and Antarctic are due to basic geography and its influence on atmospheric and oceanic circulation. The Arctic is an ocean basin surrounded largely by mountainous continental land

masses, and Antarctica is a continent surrounded by ocean. In the Arctic, sea ice extent is limited by the surrounding land masses. In the Southern Ocean winter, sea ice can expand freely into the surrounding ocean, with its southern boundary set by the coastline of Antarctica. Because Antarctic sea ice forms at latitudes further from the South Pole (and closer to the equator), less ice survives the summer. Sea ice extent in both poles changes seasonally; however, longer-term variability in summer and winter ice extent is different in each hemisphere, due in part to these basic geographical differences.

Sea ice in the Antarctic showed a slight increase in overall extent from 1979 to 2014, although some areas, such as that to the west of the Antarctic Peninsula, experienced a decrease. Short-term trends in the Southern Ocean, such as those observed, can readily occur from natural variability of the atmosphere, ocean, and sea ice system. Changes in surface wind patterns around the continent contributed to the Antarctic pattern of sea ice change; ocean factors such as the addition of cool fresh water from melting ice shelves may also have played a role. However, after 2014, Antarctic ice extent began to decline, reaching a record low (within the 40 years of satellite data) in 2017, and remaining low in the following two years.

****13. How does climate change affect the strength and frequency of floods, droughts, hurricanes, and tornadoes?****

Earth's lower atmosphere is becoming warmer and moister as a result of human-caused greenhouse gas emissions. This gives the potential for more energy for storms and certain extreme weather events. Consistent with theoretical expectations, the types of events most closely related to temperature, such as heatwaves and extremely hot days, are becoming more likely. Heavy rainfall and snowfall events (which increase the risk of flooding) are also generally becoming more frequent.

As Earth's climate has warmed, more frequent and more intense weather events have both been observed around the world. Scientists typically identify these weather events as "extreme" if they are unlike 90% or 95% of similar weather events that happened before in the same region. Many factors contribute to any individual extreme weather event—including patterns of natural climate variability, such as El Niño and La Niña—making it challenging to attribute any particular

extreme event to human-caused climate change. However, studies can show whether the warming climate made an event more severe or more likely to happen.

A warming climate can contribute to the intensity of heatwaves by increasing the chances of very hot days and nights. Climate warming also increases evaporation on land, which can worsen drought and create conditions more prone to wildfire and a longer wildfire season. A warming atmosphere is also associated with heavier precipitation events (rain and snowstorms) through increases in the air's capacity to hold moisture. El Niño events favour drought in many tropical and subtropical land areas, while La Niña events promote wetter conditions in many places. These short-term and regional variations are expected to become more extreme in a warming climate.

Earth's warmer and moister atmosphere and warmer oceans make it likely that the strongest hurricanes will be more intense, produce more rainfall, affect new areas, and possibly be larger and longer-lived. This is supported by available observational evidence in the North Atlantic. In addition, sea level rise increases the amount of seawater that is pushed on to shore during coastal storms, which, along with more rainfall produced by the storms, can result in more destructive

storm surges and flooding. While global warming is likely making hurricanes more intense, the change in the number of hurricanes each year is quite uncertain. This remains a subject of ongoing research.

Some conditions favourable for strong thunderstorms that spawn tornadoes are expected to increase with warming, but uncertainty exists in other factors that affect tornado formation, such as changes in the vertical and horizontal variations of winds.

****14. How fast is sea level rising?****

Long-term measurements of tide gauges and recent satellite data show that global sea level is rising, with the best estimate of the rate of global-average rise over the last decade being 3.6 mm per year (0.14 inches per year). The rate of sea level rise has increased since measurements using altimetry from space were started in 1992;

the dominant factor in global-average sea level rise since 1970 is human-caused warming. The overall observed rise since 1902 is about 16 cm (6 inches).

This sea level rise has been driven by expansion of water volume as the ocean warms, melting of mountain glaciers in all regions of the world, and mass losses from the Greenland and Antarctic ice sheets. All of these result from a warming climate. Fluctuations in sea level also occur due to changes in the amounts of water stored on land. The amount of sea level change experienced at any given location also depends on a variety of other factors, including whether regional geological processes and rebound of the land weighted down by previous ice sheets are causing the land itself to rise or sink, and whether changes in winds and currents are piling ocean water against some coasts or moving water away.

The effects of rising sea level are felt most acutely in the increased frequency and intensity of occasional storm surges. If CO₂ and other greenhouse gases continue to increase on their current trajectories, it is projected that sea level may rise, at minimum, by a further 0.4 to 0.8 m (1.3 to 2.6 feet) by 2100, although future ice sheet melt could make these values considerably higher. Moreover, rising sea

levels will not stop in 2100; sea levels will be much higher in the following centuries as the sea continues to take up heat and glaciers continue to retreat

****15. What is ocean acidification and why does it matter?****

Direct observations of ocean chemistry have shown that the chemical balance of seawater has shifted to a more acidic state (lower pH). Some marine organisms (such as corals and some shellfish) have shells composed of calcium carbonate, which dissolves more readily in acid. As the acidity of seawater increases, it becomes more difficult for these organisms to form or maintain their shells.

CO₂ dissolves in water to form a weak acid, and the oceans have absorbed about a third of the CO₂ resulting from human activities, leading to a steady decrease in ocean pH levels. With increasing atmospheric CO₂, this chemical balance will change even more during the next century. Laboratory and other experiments show that under high CO₂ and in more acidic waters, some marine species have misshapen shells and lower growth rates, although the effect varies among species. Acidification also alters the cycling of nutrients and many other elements and compounds in the ocean, and it is likely to shift the competitive advantage among

species, with as-yet-to-be-determined impacts on marine ecosystems and the food web.

****16. How confident are scientists that Earth will warm further over the coming century?***

Very confident. If emissions continue on their present trajectory, without either technological or regulatory abatement, then warming of 2.6 to 4.8 °C (4.7 to 8.6 °F) in addition to that which has already occurred would be expected during the 21st century.

Warming due to the addition of large amounts of greenhouse gases to the atmosphere can be understood in terms of very basic properties of greenhouse gases. It will in turn lead to many changes in natural climate processes, with a net effect of amplifying the warming. The size of the warming that will be experienced depends largely on the amount of greenhouse gases accumulating in the

atmosphere and hence on the trajectory of emissions. If the total cumulative emissions since 1875 are kept below about 900 gigatonnes (900 billion tonnes) of carbon, then there is a two-thirds chance of keeping the rise in global average temperature since the pre-industrial period below 2 °C (3.6 °F). However, two-thirds of this amount has already been emitted. A target of keeping global average temperature rise below 1.5 °C (2.7 °F) would allow for even less total cumulative emissions since 1875.

The most important amplifying feedback is caused by water vapour, which is a potent greenhouse gas. As CO₂ increases and warms the atmosphere, the warmer air can hold more moisture and trap more heat in the lower atmosphere. Also, as Arctic sea ice and glaciers melt, more sunlight is absorbed into the darker underlying land and ocean surfaces, causing further warming and further melting of ice and snow. The biggest uncertainty in our understanding of feedbacks relates to clouds (which can have both positive and negative feedbacks), and how the properties of clouds will change in response to climate change

****17. Are climate changes of a few degrees a cause for concern?****

Yes. Even though an increase of a few degrees in global average temperature does not sound like much, global average temperature during the last ice age was only about 4 to 5 °C (7 to 9 °F) colder than now. Global warming of just a few degrees will be associated with widespread changes in regional and local temperature and precipitation as well as with increases in some types of extreme weather events. These and other changes (such as sea level rise and storm surge) will have serious impacts on human societies and the natural world.

Both theory and direct observations have confirmed that global warming is associated with greater warming over land than oceans, moistening of the atmosphere, shifts in regional precipitation patterns, increases in extreme weather events, ocean acidification, melting glaciers, and rising sea levels (which increases the risk of coastal inundation and storm surge). Already, record high temperatures are on average significantly outpacing record low temperatures, wet areas are becoming wetter as dry areas are becoming drier, heavy rainstorms have become heavier, and snowpacks (an important source of freshwater for many regions) are decreasing.

These impacts are expected to increase with greater warming and will threaten food production, freshwater supplies, coastal infrastructure, and especially the welfare of the huge population currently living in low-lying areas. Even though certain regions may realise some local benefit from the warming, the long-term consequences overall will be disruptive.

****18. What are scientists doing to address key uncertainties in our understanding of the climate system?****

Science is a continual process of observation, understanding, modelling, testing, and prediction. The prediction of a long-term trend in global warming from increasing greenhouse gases is robust and has been confirmed by a growing body of evidence. Nevertheless, understanding of certain aspects of climate change remains incomplete. Examples include natural climate variations on decadal-to-centennial timescales and regional-to-local spatial scales and cloud responses to climate change, which are all areas of active research.

****19.** Are disaster scenarios about tipping points like “turning off the Gulf Stream” and release of methane from the Arctic a cause for concern?

Results from the best available climate models do not predict an abrupt change in (or collapse of) the Atlantic Meridional Overturning Circulation, which includes the Gulf Stream, in the near future. However, this and other potential high-risk abrupt changes, like the release of methane and carbon dioxide from thawing permafrost, remain active areas of scientific research. Some abrupt changes are already underway, such as the decrease in Arctic sea ice extent, and as warming increases, the possibility of other major abrupt changes cannot be ruled out.

The composition of the atmosphere is changing towards conditions that have not been experienced for millions of years, so we are headed for unknown territory, and uncertainty is large. The climate system involves many competing processes that could switch the climate into a different state once a threshold has been exceeded.

If emissions of CO₂ stopped altogether, it would take many thousands of years for atmospheric CO₂ to return to “pre-industrial” levels due to its very slow transfer to the deep ocean and ultimate burial in ocean sediments. Surface temperatures would

stay elevated for at least a thousand years, implying a long-term commitment to a warmer planet due to past and current emissions. Sea level would likely continue to rise for many centuries even after temperature stopped increasing. Significant cooling would be required to reverse melting of glaciers and the Greenland ice sheet, which formed during past cold climates. The current CO₂-induced warming of Earth is therefore essentially irreversible on human timescales. The amount and rate of further warming will depend almost entirely on how much more CO₂ humankind emits.

Scenarios of future climate change increasingly assume the use of technologies that can remove greenhouse gases from the atmosphere. In such “negative emissions” scenarios, it is assumed that at some point in the future, widespread effort will be undertaken that utilizes such technologies to remove CO₂ from the atmosphere and lower its atmospheric concentration, thereby starting to reverse CO₂-driven warming on longer timescales.

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